Roll Number: _____ Name_____ Group_____

***THAPAR INSTITUTE OF ENGINEERING &TECHNOLOGY, PATIALA***

***Department of Computer Science & Engineering***

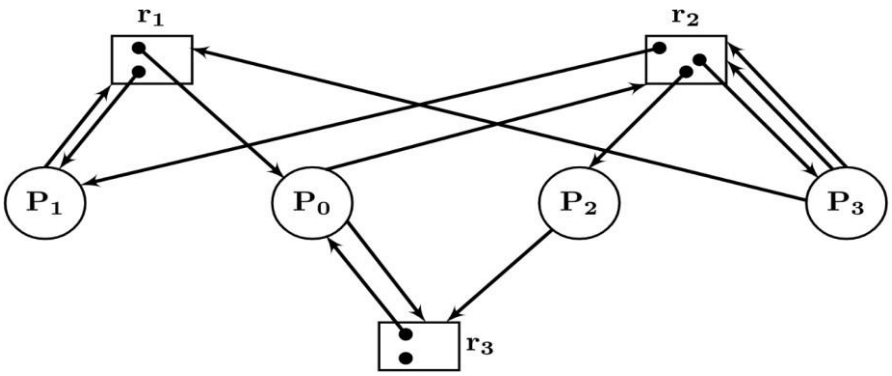
***Operating System (UCS303) - Sessional-1***

***Date: 18/09/2023 & Time: 5:45*** ***MM: 10 & MT: 12 Min***

***Note: All questions are compulsory, write correct option in the provided table.***

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

| | |
|---|---|
| 1. | In an operating system, processes that are terminated but, for some reason must have its task structure in the process table are referred as _____<br>a) Infinite Process b) Zombie Process<br>c) Orphan Process d) None of the above |
| 2. | In which of following states process resides inside the secondary memory?<br>a) New, Suspend Ready b) Running, and Waiting<br>c) Ready, Suspend Ready d) Running, and Suspend Wait |
| 3. | Predict the CPU burst time of 2nd process using exponential averaging, where predicted burst time of 1st process is 8 units and actual burst time for the 1st process is 3 units (take $\alpha = 0.8$).<br>a) 5 b) 4<br>c) 6 d) 3 |
| 4. | Consider a scenario in which a CPU-bound process requires a burst time of 40 seconds. The Multilevel feedback queue scheduling algorithm is used, and the queue time quantum is 2 seconds, and in each level, it is incremented by 5 seconds. How many times process will be interrupted before completing its execution?<br>a) 3 b) 4<br>c) 6 d) 5 |
| 5. | A system is having 3 user processes P1, P2 and P3, where P1 requires 4 units of resource R, P2 requires 4 units of resource R, P3 requires 6 units of resource R. The minimum number of units of R that ensures there is no deadlock in system is _____?<br>a) 12 b) 11<br>c) 13 d) 14 |
| 6. | A system state is represented by the resource allocation graph given below analyze the system and tell which one is true? |




a) Safe sequence exists
b) Safe sequence doesn't exist
c) <P0 P1 P3 P2> is safe sequence
d) <P3 P1 P0 P2> is a safe sequence

7. In a dual mode operating system, _____ is responsible for managing memory allocation and process scheduling having mode bit value _____?

a) User mode, 1
b) Kernel mode, 0
c) Supervisor mode, 1
d) Protected mode, 1

8. How many times will the "Welcome to OS Quiz" be printed?

```
void main (){
    if (fork() || fork()){
        fork();
        printf("Welcome to OS Quiz \n"); } }
```

a) 5
b) 6
c) 4
d) 8

9. In _____user services and kernel, services are kept in separate address spaces.

a) Microkernel kernel
b) Monolithic kernel
c) Hybrid kernel
d) None of the above

10. Suppose that the following processes arrive for execution at indicated times. Each process will run for the amount of time listed. What is the completion time of the process P2, when non-preemptive SJF scheduling algorithm's is used?

| Process | Arrival Time | Burst Time |
|---|---|---|
| P1 | 0 | 3 |
| P2 | 4 | 4 |
| P3 | 4 | 1 |

a) 4
b) 5
c) 9
d) 8

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***Thapar Institute of Engineering & Technology, Patiala***

***DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING***

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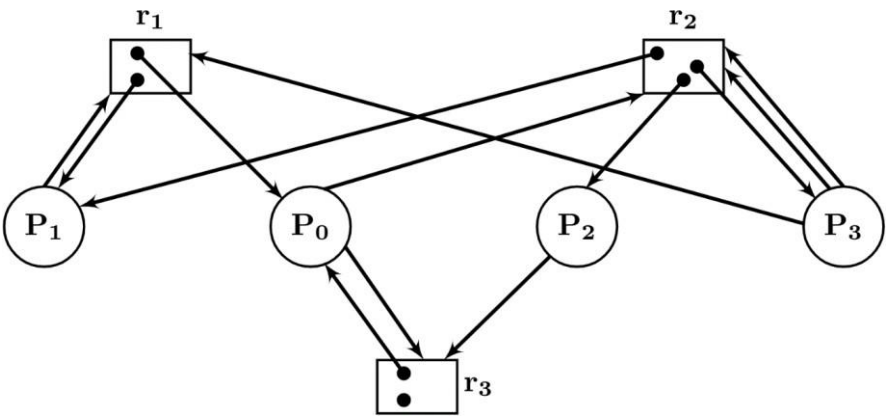
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

| | |
|---|---|
| 1. | If a parent did not invoke wait() and instead terminated, thereby leaving its child processes as it is, the child process is known as _____<br>a) Infinite Process b) Zombie Process<br>c) Orphan Process d) None of the above |
| 2. | In which of following states process resides inside the main memory?<br>a) New, Ready, and Running b) Ready, Running, and Waiting<br>c) Ready, Suspend Ready and Waiting d) New, Running, and Waiting |
| 3. | Predict the CPU burst time of 2nd process using exponential averaging, where predicted burst time of 1st process is 4 units and actual burst time for the 1st process is 1 unit (take $\alpha = 0.8$).<br>a) 1.4 b) 1.8<br>c) 1.6 d) 2.0 |
| 4. | Consider a scenario in which a CPU-bound process requires a burst time of 32 seconds. The Multilevel Feedback Queue scheduling algorithm is used, and the queue time quantum is 3 seconds, and in each level, it is incremented by 7 seconds. In which queue will the process terminate the execution?<br>a) 3 b) 4<br>c) 6 d) 5 |
| 5. | A system is having 3 user processes P1, P2 and P3 where P1 requires 4 units of resource R, P2 requires 5 units of resource R, P3 requires 6 units of resource R. The minimum number of units of R that ensures there is no deadlock in system is _____?<br>a) 16 b) 15<br>c) 12 d) 13 |
| 6. | A system state is represented by the resource allocation graph given below analyze the system and tell which one is correct? |




a) P1 will be the last process in safe sequence
b) System will be in deadlock free state
c) Safe sequence doesn't exist
d) P0 will be the last process in safe sequence

7. The primary responsibility of the operating system is _____in kernel mode having mode bit value as_____?

a) Running user applications, 0
b) Managing hardware resources, 0
c) Providing a graphical user interface, 1
d) Running device drivers, 1

8. How many times will the "Welcome to OS Quiz" be printed?

```
void main (){
    if (fork() && fork()){
        fork();
        printf("Welcome to OS Quiz \n");}}
```

a) 5
b) 4
c) 2
d) 8

9. A _____is a type of kernel in operating systems where the entire operating system works in the kernel space.

a) Microkernel kernel
b) Monolithic kernel
c) Hybrid kernel
d) None of the above

10. Suppose that the following processes arrive for execution at indicated times. Each process will run for the amount of time listed. What is the completion time of the process P3, when non-preemptive SJF scheduling algorithm's is used?

| Process | Arrival Time | Burst Time |
|---|---|---|
| P1 | 0 | 3 |
| P2 | 4 | 4 |
| P3 | 4 | 1 |

a) 4
b) 5
c) 9
d) 8

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***Thapar Institute of Engineering & Technology, Patiala***

***Department of Computer Science & Engineering***

***OPERATING SYSTEM (UCS303) – SESSIONAL-1***

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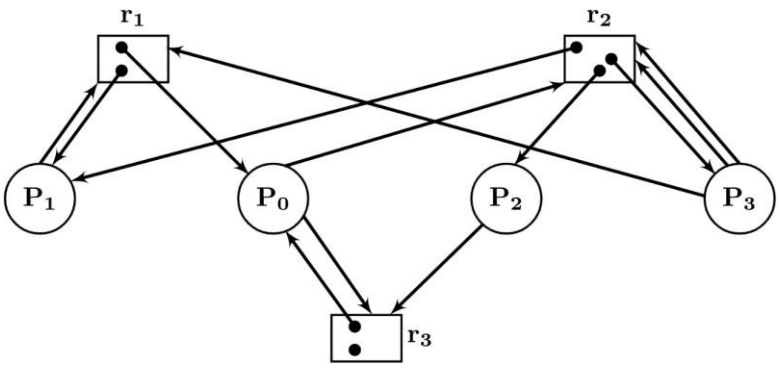
***Note: All questions are compulsory, write correct option in the provided table.***

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

| | |
|---|---|
| 1. | Consider the following program:<br>main() { if(fork()>0)<br>sleep(100); }<br>a) Infinite Process b) Zombie Process<br>c) Orphan Process d) None of the above |
| 2. | In which of following state sequence process resides inside the main memory, secondary memory and main memory.<br>a) Ready, Running, Suspend Ready b) Running, Waiting, Suspend Wait<br>c) Ready, Suspend Ready, Waiting d) Running, Suspend Wait, Suspend Running |
| 3. | Predict the CPU burst time of 2nd process using exponential averaging, where predicted burst time of 1st process is 10 units and actual burst time for the 1st process is 6 units (take $\alpha = 0.5$).<br>a) 8 b) 4<br>c) 6 d) 5 |
| 4. | Consider a scenario in which a CPU-bound process requires a burst time of 32 seconds. The Multilevel Feedback Queue scheduling algorithm is used, and the queue time quantum is 3 seconds, and in each level, it is incremented by 7 seconds. How many times process will be interrupted before completing its execution?<br>a) 3 b) 4<br>c) 6 d) 5 |
| 5. | A system is having 3 user processes P1, P2 and P3 where P1 requires 3 units of resource R, P2 requires 4 units of resource R, P3 requires 5 units of resource R. Maximum number of units of resource R that ensures deadlock in system is_____?<br>a) 12 b) 9<br>c) 10 d) 8 |
| 6. | A system state is represented by the resource allocation graph given below analyze the system and tell which one is true |




a) P1 will execute in last
b) P2 will execute in last
c) P3 will execute in last
d) P0 will execute in last

7. _____of operation in a dual mode system allows unrestricted access to system resources and hardware having mode bit value as _____?

a) User mode, 0
b) Kernel mode, 0
c) Supervisor mode, 1
d) Protected mode, 1

8. How many times will the "Welcome to OS Quiz" be printed?

```
void main (){
    fork();
    if (fork() || fork()){
        printf("Welcome to OS Quiz\n");}}
```

a) 5
b) 6
c) 4
d) 8

9. A _____is a single large program that contains all the system calls and device drivers.

a) Microkernel kernel
b) Monolithic kernel
c) Hybrid kernel
d) None of the above

10. Suppose that the following processes arrive for execution at indicated times. Each process will run for the amount of time listed. What is the average waiting time for whole system with three processes given here, when non-preemptive SJF scheduling algorithm's is used?

| Process | Arrival Time | Burst Time |
|---|---|---|
| P1 | 0 | 3 |
| P2 | 4 | 4 |
| P3 | 4 | 1 |

a) 0.25
b) 0.33
c) 0.50
d) 0.75